Experiment No-15 : Decision Tree

Aim

To write a Python program to implement a Decision Tree classifier in Artificial Intelligence using the Iris dataset.

Programming Code:

```
from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier, export_text
from sklearn.metrics import accuracy_score

# Load the Iris dataset
iris = load_iris()
X, y = iris.data, iris.target

# Split into training and testing sets
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, random_state=42)

# Build and train the decision tree
model = DecisionTreeClassifier(criterion="entropy", max_depth=3, random_state=42)
model.fit(X_train, y_train)

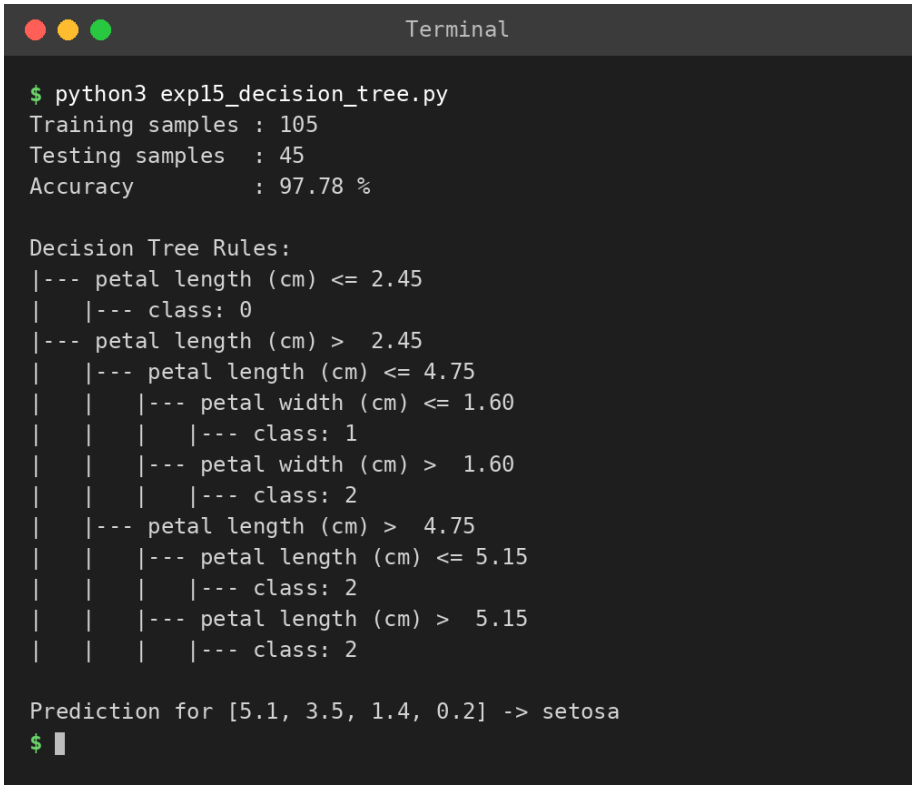
# Test the model
y_pred = model.predict(X_test)
accuracy = accuracy_score(y_test, y_pred)

print("Training samples :", len(X_train))
print("Testing samples  :", len(X_test))
print("Accuracy         :", round(accuracy * 100, 2), "%")

print("\nDecision Tree Rules:")
print(export_text(model, feature_names=iris.feature_names))

# Predict for a new flower
sample = [[5.1, 3.5, 1.4, 0.2]]
prediction = model.predict(sample)[0]
print("Prediction for", sample[0], "->", iris.target_names[prediction])
```

OUTPUT:



Expected Output

- The dataset is split 70:30, giving 105 training samples and 45 testing samples.
- The accuracy on the test data is displayed as 97.78 %, meaning 44 of the 45 test samples are classified correctly.
- The tree rules are printed. The root splits on petal length <= 2.45 and gives class 0 (setosa).
- The remaining samples are split further using petal length (4.75) and petal width (1.60) to separate class 1 (versicolor) and class 2 (virginica).
- The prediction for the new sample [5.1, 3.5, 1.4, 0.2] is setosa.

Result

Thus, the Decision Tree algorithm was implemented successfully in Python. The model learned if-else rules from the training data and classified the Iris flowers with an accuracy of 97.78 %.